

PID (Proportional + Integral + Derivative) Controller

Action: Combines P for speed, I for zero steady-state error, D for damping.

Features:

- Fast, stable, accurate
- Most widely used
- Slightly more complex to tune

IoT & Robotics Examples:

Application	How Used
Industrial robotic arms	Precise position control without overshoot
Drone autopilot (altitude & attitude)	Stable flight with exact position and minimal oscillation
Smart motor drivers in IoT	Fast, accurate, and smooth speed control
Autonomous vehicles	Steering and speed control for safe navigation

PID = all-rounder, common in automation, IoT, robotics.

PD (Proportional + Derivative) Controller

Action: Combines P for speed, D for damping.

Features:

- Reduces overshoot
- Faster than P alone
- Does **not** remove steady-state error

IoT & Robotics Examples:

Application	How Used
Robotic arm trajectory control	Fast motion with less oscillation
Camera gimbal stabilization	Smooth tracking with minimal shake
Drone pitch/roll control	Quick response with reduced wobble

PD = speed + stability, good for transient response.

PI (Proportional + Integral) Controller

Action: Combines P for speed, I for zero steady-state error.

Features:

- Faster than I alone
- Eliminates steady-state error
- Slight overshoot

IoT & Robotics Examples:

Application	How Used
Smart HVAC systems	Quickly reaches set temperature and maintains it
Motor speed control in robots	Fast response + exact speed tracking
IoT water level control	Fast and accurate water level regulation

PI is very popular in industry and IoT, balances speed and accuracy.

D (Derivative) Controller

Action: Output $\propto$ rate of change of error.

Features:

- Improves stability
- Reduces overshoot
- Does **not** fix steady-state error

IoT & Robotics Examples:

Application	How Used
Robotic arm motion smoothing	Dampens rapid swings in joint movement
Drone stabilization	Reduces oscillations during fast maneuvers
Self-balancing robot (like Segway)	Prevents tipping by responding to tilt rate

D is like a predictor, improving smoothness and damping.

I (Integral) Controller

Action: Output $\propto$ accumulated error over time.

Features:

- Eliminates steady-state error
- Slower response
- Can reduce stability

IoT & Robotics Examples:

Application	How Used
Smart water level tank	Keeps water at exact level even if inflow changes
Drone altitude hold	Accumulates tiny altitude errors to maintain exact height
Battery charge control in IoT devices	Ensures precise voltage/current target is reached

I is used when accuracy matters, steady-state error must be zero.

P (Proportional) Controller

Action: Output $\propto$ current error.

Features:

- Fast response
- Leaves steady-state error
- Simple

IoT & Robotics Examples:

Application	How Used
Room temperature control (smart thermostat)	Heater/fan adjusts proportional to temperature difference
Light intensity control (smart lighting)	LED brightness adjusts to difference from desired lux level
Motor speed in simple robots	Motor voltage changes proportional to speed error for small movements

P is for simple, fast correction, not precise final value.

Q. In an IoT water tank system, the water level must be maintained accurately despite varying inflow. Which controller(s) would you use and why?

Q. A robotic arm needs to reach a target position quickly but without overshoot. Suggest an appropriate controller type and explain its effect on speed and damping.

Q. A drone experiences oscillations in flight. Which controller or combination of controllers would you recommend to reduce oscillations and maintain precise altitude? Explain why.

Q. Give one example of an IoT or robotics application where PID control is essential.

Q. Explain with examples how PID controllers are applied in IoT and robotics to achieve fast, stable, and accurate control. Include applications such as drones, robotic arms, and industrial automation.

Q. A smart robotic system uses temperature control, motor speed control, and stabilization. Identify which type of controller (P, I, D, PI, PD, PID) is most suitable for each task and justify your choices.